Unit 1 Progress Check: FRQ Part C

This question involves processing an email address to create three usernames.

The email address is stored in a String variable named email, which consists of a first name, followed by ".", followed by a last name, followed by "@", followed by a domain name. Assume that the email address is stored in this format and the last name has at least three characters.

The usernames will be formatted as follows.

- Username 1 will consist of the first name, followed by "_", followed by the last name.
- Username 2 will consist of the first initial of the first name, followed by "_", followed by the last name.
- Username 3 will consist of a randomly-selected character from the first name , followed by "_", followed by a sequence of three consecutive characters selected from a random position in the last name.

Example 1

Suppose email is equal to "joseph.martin@school.org".

- Username 1 will be "joseph_martin".
- Username 2 will be "j_martin".
- Possible values of Username 3 include "s_art" and "p_tin".

Example 2

Suppose email is equal to "virginia.farmingham@school.net".

- Username 1 will be "virginia_farmingham".
- Username 2 will be "v_farmingham".
- Possible values of Username 3 include "g_ham" and "i_arm".

1. Write a statement that uses the String method indexOf to find the index of the first "." in email and assign it to a variable. Write another statement that uses the String method indexOf to find the index of "@" and assign it to a variable.

2. Write a statement that uses the String method substring to extract the first name from email and assign it to a variable. Write another statement that uses the String method substring to extract the last name from email and assign it to a variable. Then use the results of those statements to create and assign the correct value to a String variable for username 1.

3. Write a statement that uses the String method substring to extract the first initial of the first name from email and assign it to a variable. Use the results of this statement and the statements you wrote in part B to create and assign the correct value to a String variable for username 2.

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4. Write a statement that uses the Math method random to generate a random number between 0 and the length of the first name. Write another statement that uses the Math method random to generate a random number between 0 and the length of the last name minus 2.

5. Write a code segment that uses the random numbers that you generated in part D to create and assign the correct value to a String variable for username 3.